Fiscal Year:	FY 2006	**Task Last Updated:**	FY 08/04/2005
PI Name:	Ceperley, David M. Ph.D.		
Project Title:	Properties of liquid helium from computer simulation		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Web Page:	http://www.physics.uiuc.edu/		
City:	Urbana	**State:**	IL
Zip Code:	61801	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	10/01/2001	**End Date:**	01/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Goldenfeld, Nigel (University of Illinois Urbana-Champaign)		
Grant/Contract No.:	NNC04GA01G		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		
Task Description:	We are conducting accurate microscopic simulations, based on realistic interatomic potentials, of bulk superfluid, superfluids in restricted geometeries,BEC of fermion gases and arrays of vortices. The aim of this research is to provide reliable, quantitative theoretical predictions which will aid in the design and interpretation of low temperature experiments performed in the microgravity facility. Considerable progress can be achieved on these fronts, owing to the possibility of performing accurate measurements in a gravity- controlled environment and making comparison with state-of-the-art numerical calculations. This research is concerned with fundamental aspects of helium and quantum fluids in general; we address outstanding problems in the current understanding of relevant phenomena such as Bose condensation, superfluidity, and phase transitions, as well as of theoretical issues such as the inference of bulk properties of matter from the study of finite clusters. The theoretical issues involved in helium systems are of direct relevance to understanding other many-body quantum systems such as correlated electronic systems		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	During the past year we have 1) completed a study of arrays of vortices, for comparision with mean field results and experiment in cold atomic gases. Results being prepared for publication. 2) finished a simulation study of 4He confined in porous Vycor, and obtained agreement with experiment of Chan on superfluid response and its dependance on 3He concentration. Publication is under review. 3) calculated probability of supercurrents in bulk 4He in order to understand experiments of M. Chan. We do not find supersolid behavior, suggesting that explanation of experiments is not with conventional mechanism. Research published in Phys. Rev. Letts.
Bibliography Type:	Description: (Last Updated: 09/11/2009)
Articles in Peer-reviewed Journals	Ceperley DM, Bernu B. " Ring exchanges and the supersolid phase of 4He." Phys Rev Lett. 2004 Oct 8;93(15):155303. PMID: 15524899 , Oct-2004
Dissertations and Theses	Hector Garcia Martin. "STATISTICAL ANALYSIS OF HIGHLY CORRELATED SYSTEMS IN ..." PhD thesis, University of Illinois Urbana-Champaign , January 2005. , Jan-2005